

# Use Case Place Order

## 1. Use Case Code

UC001

## 2. Brief Description

The use case “Place Order” describes the interaction between the customer and the AIMS Software. It is invoked when the customer wishes to place an order after the Customer has finished selecting products.

## 3. Actor(s)

There is only 1 Actor interacting with this use case: Customer.

## 4. Precondition(s)

The use case can only be invoked if the following condition is satisfied:

- Customer has added at least 1 product in cart.

## 5. Basic Flow of Events

Step 1. Customer places order

Step 2. AIMS Software checks for stock availability

Step 3. AIMS Software displays the delivery form and prompts the customer to set up delivery information

Step 4. Customer enters delivery information and submits

Step 5. AIMS Software checks for missing required fields and input validity

Step 6. AIMS Software calculates and displays the delivery fee

Step 7. AIMS Software displays and temporarily saves invoice information

Step 8. Customer requests to pay for the order

Step 9. AIMS Software calls the use case “Pay Order”

Step 10. AIMS Software puts the order in pending state

Step 11. AIMS Software displays general order information and transaction information to screen

Step 12. AIMS Software sends invoice and payment transaction information to customer's email

Step 13. AIMS Software records payment transaction information and successfully paid order

## 6. Alternative Flows

| No | Location | Condition                                               | Action                                                                                                               | Resume Location |
|----|----------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------|
| 1  | Step 2   | Stock is insufficient                                   | AIMS software prompts the customer to update the cart and displays the available quantity for each affected product. | Use case ends   |
| 2  | Step 5   | Missing required fields or invalid inputs               | AIMS Software prompts the customer to fill in missing fields or reenter                                              | Step 4          |
| 3  | Step 7   | Customer requests to adjust the delivery method         | AIMS Software displays the delivery form and prompts the customer to reenter                                         | Step 4          |
| 4  | Step 7   | Customer requests to adjust product list                | AIMS Software displays cart and prompts the customer to update the cart                                              | Step 1          |
| 5  | Any step | Customer requests to go back to step N                  | AIMS Software redirects to step N                                                                                    | Step N          |
| 6  | Any step | Customer exits the software or requests to cancel order | AIMS Software cancels the ordering process                                                                           | Use case ends   |
| 7  | Step 9   | Order payment is not successful                         | AIMS Software displays notification of failed transaction                                                            | Use case ends   |

*Table 1-Alternative flows of events for UC Place Order*

## 7. Input data

| No | Data Fields                    | Description            | Mandatory | Valid Condition                                                                                         | Example                                                                            |
|----|--------------------------------|------------------------|-----------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1  | Customer name                  |                        | Yes       |                                                                                                         | Nguyễn Văn A                                                                       |
| 2  | Phone number                   |                        | Yes       | - 10 digits<br>- Starts with 0<br>- May contain separators such as '-', '.', or space in between digits | 0123456789<br>0123.456.789<br>0123-456-789<br>0123 456 789<br>012.34.56.789<br>... |
| 3  | Province/City (Tỉnh/Thành phố) | Selectable from a list | Yes       |                                                                                                         | Hà Nội<br>Nghệ An<br>...                                                           |
| 4  | Ward/Commune (Phường/Xã)       | Selectable from a list | Yes       |                                                                                                         | Phường Hà Đông<br>Xã An Khánh<br>...                                               |

|   |                          |  |     |  |                                   |
|---|--------------------------|--|-----|--|-----------------------------------|
| 5 | Address                  |  | Yes |  | 199 Chiến<br>Thắng, Triều<br>Khúc |
| 6 | Delivery<br>instructions |  | No  |  |                                   |

*Table 2-Input data of Delivery Information*

## 8. Output data

| No | Data Fields                     | Description                                                   | Display Format                                      | Example                                                                     |
|----|---------------------------------|---------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------|
| 1  | Product type +<br>title         | The type and the title<br>of the product                      | [Type] [Title]                                      | Book Data<br>Structures &<br>Algorithms<br>DVD Ice Age<br>Collection<br>... |
| 2  | Quantity                        | The number of the<br>corresponding product<br>to be purchased | Positive integer                                    | 3                                                                           |
| 3  | Price                           | Price of the product                                          | Positive integer<br>Includes comma for<br>thousands | 99,000                                                                      |
| 4  | Amount                          | Total price for the<br>product                                | Positive integer<br>Includes comma for<br>thousands | 297,000                                                                     |
| 5  | Total product<br>price          | Total amount of all<br>products                               | Positive integer<br>Includes comma for<br>thousands | 1,423,000                                                                   |
| 6  | Total product<br>price with VAT | Total product price +<br>VAT                                  | Positive integer<br>Includes comma for<br>thousands | 1,585,300                                                                   |
| 7  | Delivery fee                    |                                                               | Positive integer<br>Includes comma for<br>thousands | 50,000                                                                      |
| 8  | Total amount                    | The amount to be<br>paid                                      | Positive integer<br>Includes comma for<br>thousands | 1,635,300                                                                   |

*Table 3-Output data of Invoice Information*

| No | Data Fields                       | Description | Display Format                                          | Example                                                                            |
|----|-----------------------------------|-------------|---------------------------------------------------------|------------------------------------------------------------------------------------|
| 1  | Customer name                     |             |                                                         | Nguyễn Văn A                                                                       |
| 2  | Phone number                      |             | 10 digits starting<br>with 0 with '.', '-', or<br>space | 0123456789<br>0123.456.789<br>0123-456-789<br>0123 456 789<br>012.34.56.789<br>... |
| 3  | Province/City<br>(Tỉnh/Thành phố) |             |                                                         | Hà Nội<br>Nghệ An<br>...                                                           |

|   |                          |                       |                                                  |                                      |
|---|--------------------------|-----------------------|--------------------------------------------------|--------------------------------------|
| 4 | Ward/Commune (Phường/Xã) |                       |                                                  | Phường Hà Đông<br>Xã An Khánh<br>... |
| 5 | Address                  |                       |                                                  | 199 Chiến Thắng,<br>Triều Khúc       |
| 6 | Delivery instructions    |                       |                                                  |                                      |
| 7 | Total amount             | The amount to be paid | Positive integer<br>Includes comma for thousands | 1,635,300                            |

*Table 4-Output data of Order Information*

| No | Data Fields          | Description | Display Format | Example    |
|----|----------------------|-------------|----------------|------------|
| 1  | Transaction ID       |             |                |            |
| 2  | Transaction Content  |             |                |            |
| 3  | Transaction Datetime |             | dd/mm/yyyy     | 15/03/2026 |

*Table 5-Output data of Transaction Information*

## 9. Postconditions

The use case can be considered successful if all the following conditions are satisfied:

- A new order is created in the system and its status is set to Pending.
- The system records payment transaction information and the successfully paid order
- Customer receives email containing invoice and payment transaction information

If the use case ends prematurely or is unsuccessful due to payment failure, the ordering process is reset and nothing else happens.